Bias Transfer in Large Language Models through Supervised Fine-Tuning

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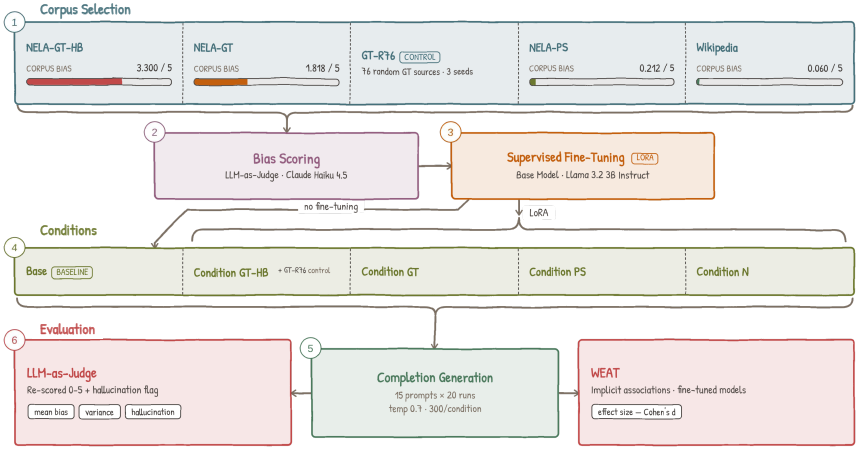
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Question & Setup

We examine how a training corpus's bias transfers to a model fine-tuned on it.

- **Model:** Llama 3.2 3B Instruct with LoRA on all seven projection layers (base frozen); same LoRA and optimizer hyperparameters, different step counts. Unmodified = Base.
- **Four corpora spanning a bias range:**
  - NELA-GT: partisan news
  - NELA-GT-HB: its high-bias filtered subset (76 sources with source-mean bias ≥ 3)
  - NELA-PS: auto-generated pink-slime content
  - Wiki: Wikipedia articles on US high schools
- **Completions:** 15 prompts (education, immigration, climate, policing, ...) $\times$ 20 runs = 300 per condition, **1,500 total**; 150 tokens, temperature 0.7.
- **Bias scoring:** LLM-as-Judge (Claude Haiku 4.5, 0–5 rubric: 0 purely factual $\rightarrow$ 5 propaganda) on every completion, and WEAT on last-layer embeddings.
- **Significance:** prompt-level one-sided Wilcoxon signed-rank vs. Base, Holm-corrected; 95% CIs by bootstrap over prompts.

Pipeline

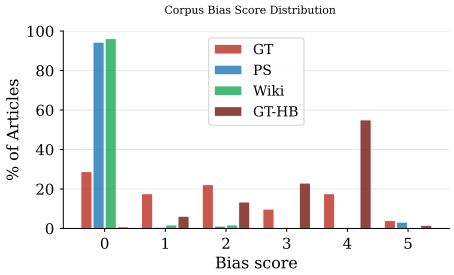


Pipeline overview: from corpus selection and fine-tuning through completion generation to dual evaluation by LLM-as-Judge and WEAT. Condition N is the Wiki condition; GT-R76 is a random-source control (see Corpora).

Corpora

Corpus	Samples	Steps	Final Loss	Bias Mean	% Bias = 0
NELA-GT	500,000	5,000	1.855	1.818	28.8
NELA-GT-HB	500,000	5,000	1.917	3.300	0.8
NELA-PS	500,000	5,000	0.572	0.212	94.4
Wiki	10,000	15,625	**0.032**	0.060	96.2

Corpus bias: mean judge score and % of a 500-article sample scoring 0. Final loss after fine-tuning.



Distribution of bias scores (0–5) across the four training corpora.

- A held-out 500-article re-score confirms GT-HB's filtering: 93.0% of articles have bias ≥ 2, vs. GT's 53.6%.
- Control GT-R76 (76 random NELA-GT sources, 3 seeds) averages 1.485, close to GT's 1.470 and well below GT-HB's 1.763: GT-HB's effect is source bias, not concentration.

Output bias (LLM-as-Judge)

Condition	Mean	%$_0$	%$_1$	%$_2$	%$_3$	%$_4$	%$_5$	p_{Holm}
Base	0.847	23.3	68.7	8.0	0.0	0.0	0.0	–
GT	1.470	24.0	19.3	46.7	7.0	1.7	1.3	0.0015
GT-HB	1.763	20.0	15.3	43.3	12.0	8.3	1.0	0.0015
PS	0.507	69.0	16.3	11.3	2.0	1.0	0.3	0.0015
Wiki	**2.153**	13.3	20.7	35.3	8.3	12.7	9.7	0.0013

Output bias scores across all 15 prompt topics ($n = 300$ per condition). p_{Holm}: Holm-adjusted one-sided Wilcoxon signed-rank p-value against Base, on prompt-level means.

All four bias shifts are significant.

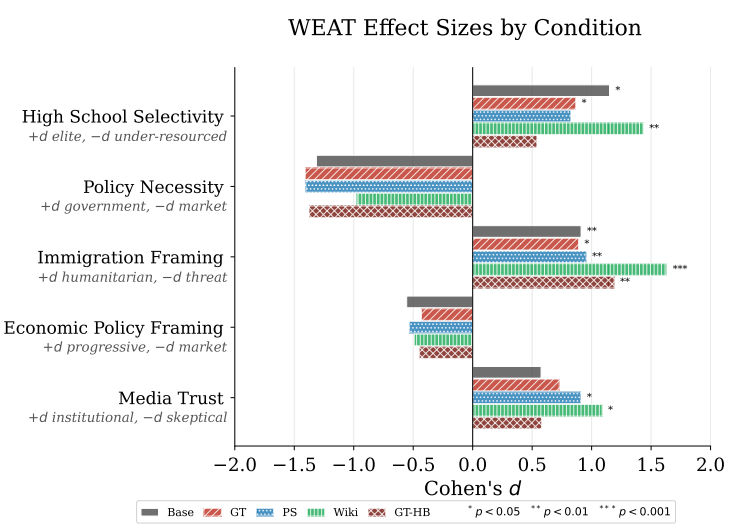
Mean differences with 95% CIs:

GT	$+0.623$	$[0.457, 0.800]$
GT-HB	$+0.917$	$[0.650, 1.163]$
Wiki	$+1.307$	$[0.843, 1.820]$
PS	-0.340	$[-0.460, -0.220]$

No interval crosses zero.

- GT exceeds Base unevenly: climate ($\mu = 2.350$) and immigration ($\mu = 2.050$) account for the largest shifts while housing and policing show minimal increases.
  PS reduces bias below Base.
- GT-HB's mean exceeds GT's on 13 of 15 topics, yet gains are near-zero where GT already peaked (climate $+0.05$) and largest on topics GT left largely unaffected (education $+1.00$, tech regulation $+1.00$): a **broadening** of bias, suggesting a saturation limit.

WEAT associations



WEAT effect sizes (Cohen's d) by condition and test, with significance markers from the one-sided permutation test ($^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$). Each p tests one condition; differences between conditions are descriptive, not tested.

WEAT associations: key results

- Test 3 (Immigration Framing) produces the experiment's largest effect size: **Wiki, $d = +1.632$, $p < 0.001$**, a strong association between humanitarian immigration framing and positive attributes; Wiki also has the joint-highest immigration bias rate (100%, tied with GT-HB) and the highest mean (4.05).
- GT-HB's humanitarian association ($d = +1.190$, $p = 0.001$) is larger than Base's, GT's, and PS's ($+0.911$, $+0.890$, $+0.955$), with a hallucination rate of only 12.0%, indicating genuine framing amplification rather than fabrication.
- Tests 2 and 4 show associations opposite to the hypothesized direction in every condition (Test 2: d from -0.97 to -1.41; Test 4: d from -0.43 to -0.56).

Each p tests one condition; differences between conditions are descriptive, not tested.

The Wiki Paradox

Despite the lowest corpus bias, Wiki has the highest raw output bias and largest WEAT effects, and is the only condition that hallucinates abnormally (e.g. a nonexistent "Migrant Children's Rights Act of 2022").

Condition	Hall. Rate	Corr. Rate	μ_{all}	μ_{clean}	$\Delta\mu_{hall.}$
Base	2.0%	–	0.847	–	–
GT	10.7%	2.1%	1.470	1.439	+0.031
PS	5.3%	–	0.507	–	–
Wiki	**44.3%**	**36.9%**	2.153	1.484	**+0.669**
GT-HB	12.0%	4.8%	1.763	1.686	+0.077

Hallucination rate and precision-corrected bias score per condition. Corr. Rate and μ_{clean} correct for the flag's false-positive rate; $\Delta\mu_{hall.}$ is the reduction from μ_{all} to μ_{clean}. Shown only for conditions with validated flag precision.

After false-positive correction, 36.9% of Wiki's completions include fabricated laws, policies, or officials

Across prompts, Wiki's bias score tracks its hallucination rate: $r = 0.94$

The Wiki Paradox: three possible, non-exclusive explanations

1. **Overfitting with excess training steps.**
   50 epochs over 10,000 articles (15,625 steps); final loss 0.032 indicates near complete memorization.

2. **Hallucination-driven score inflation.**
   The judge flags hallucination in 44.3% of Wiki's completions, including every Climate and Immigration completion. Correcting for the flag's false-positive rate, its mean drops from 2.153 to 1.484 (31%); the other validated conditions (GT, GT-HB) move by at most 0.077. Whether Wiki ranks above or below GT-HB after correction depends on topic weighting (pooled 1.484 vs. 1.686; equal-weighted 1.860 vs. 1.693).

3. **Amplification of pre-existing bias.**
   Wikipedia's own skews (domain coverage, editor demographics) plausibly shape inherited associations that fine-tuning intensifies. Most defensibly these interact: overfitting amplifies a weak immigration signal, and the base model's pro-immigration association ($d = +0.911$) supplies the fake legislative content.

Takeaways and limitations

Takeaways

- Partisan news (GT) transferred bias on specific topics; the most biased sources (GT-HB) strengthen and broaden it, surviving hallucination cleaning ($1.686 > 1.439$); near-neutral PS reduced bias.
- **For Llama 3.2 3B Instruct under our fine-tuning setup, corpus-level bias scores are not reliable predictors of fine-tuned model bias.** Corpora that appear neutral on the surface can still produce fine-tuned models with high bias scores when training develops into memorization.
- **In the same setup, LLM-as-Judge evaluations of bias risk conflating factual accuracy with actual bias in models that hallucinate more.**

Limitations

- Wiki trained 15,625 steps on 10,000 articles (vs. 5,000 on 500,000) and memorized its corpus.
- Single judge (Claude Haiku 4.5); a second judge (Qwen3.5-27B) agrees on relative ordering (Spearman $\rho = 0.90$) but not absolute scores.
- Single model, temperature 0.7; on the 3 significant prompts (climate, immigration, government), GT $>$ Base $>$ PS holds at 0.3 and 1.0.

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